

Comparative Analysis of Haemostatic Proteinases: Structural Aspects of Thrombin, Factor Xa, Factor IXa and Protein C

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Introduction

Haemostasis is the delicate balance between the processes of coagulation, anticoagulation and fibrinolysis. These processes are governed by activation cascades involving trypsin-like serine proteinases, of which the coagulatory cascade is perhaps the best characterized (1). Vascular rupture causes exposure to the blood of the membrane bound tissue factor to form a complex with factor VIIa (2), which then activates factor Xa (extrinsic activation). In turn, factor Xa, which binds to the phospholipid membrane in the presence of calcium, catalyses the formation of active α -thrombin from its zymogen precursor prothrombin. Factor Xa achieves full activity against prothrombin only in the presence of factor Va, forming the prothrombinase complex (3). The thrombin-catalysed activation of the coagulation cofactors factor Va and factor VIIIa provides a positive feedback mechanism that causes a burst of thrombin production. Factor VIIIa forms a complex with factor IXa, allowing the generation of large amounts of active factor Xa (intrinsic activation). Thrombin also initiates the anticoagulatory cascade: in the presence of thrombomodulin, the activity of thrombin is switched from the procoagulatory conversion of fibrinogen to fibrin to the activation of the anticoagulant protein C (4). Protein C forms a complex with its cofactor protein S, and proceeds to selectively inactivate factors Va and VIIIa.

Despite their highly similar architectures, the procoagulant proteinases (thrombin, factor Xa, factor IXa and factor VIIa), the anticoagulant protein C, and the fibrinolytic enzymes

(tissue-type plasminogen activator and plasminogen) display major differences in their specificities and activities. The large body of information that has accumulated on the pathology, biochemistry and molecular biology of blood coagulation has now been supplemented by three dimensional structural data from X-ray crystallography and NMR (5). Although the structure of α -thrombin (6,7) has explained many aspects of the specificity of thrombin -reviewed in (8,9), more recent data show that thrombin is a dynamic molecule whose activity is allosterically regulated. In this review, we shall focus on the novel aspects of thrombin structure, and compare and contrast structural aspects of the plasma proteinases factor Xa (10,11), factor IXa (12) and protein C (13). Other articles in this volume will discuss the tissue factor:factor VIIa complex (14,15) and serpins (16).

The catalytic units of the haemostatic proteinases share the fold of the trypsin-like serine proteinases, i.e. two interconnected six-stranded antiparallel β -barrel domains, with the catalytic residues located at the junction between them (see Fig. 1). This common architecture permits the use of the chymotrypsinogen numbering system for residues of the catalytic domains, which facilitates comparison of the various factors. Thus, the catalytic residues are denoted by Ser195, His57 and Asp102 (corresponding in human thrombin to residues Ser525, His363 and Asp419, in factor Xa to residues Ser379, His236 and Asp282, in factor IXa to residues Ser365, His221 and Asp269, and in protein C to residues Ser360, His211 and Asp257, respectively). Similar to trypsin, the catalytic domains of factor Xa, factor IXa and protein C all possess a Ca^{2+} binding site formed by acidic residues Glu70 (Asp in factor Xa), Glu77 and Glu80 of the 70-80 loop; in thrombin, this location is occupied by the amino group of Lys70, yielding a stability independent of external calcium levels (see Figs 1 and 4). The individual proteinases are distinguished by surface loops of varying lengths connecting the common secondary structure elements; those bordering the active site cleft, listed in Table 1, are responsible for interacting with substrates and inhibitors.

In addition to the common scaffold of the catalytic domains, the haemostatic proteinases share a number of modu-

Dedication. This paper is dedicated to Professor Dr. Robert Huber on the occasion of his 60th birthday

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Table 1. Sequence comparisons of the interacting surfaces of the coagulation proteinases

Chg	36	37	a	38	39	40	41	42		60	a	b	c	d	e	f	g	h	i	61
FII	K	S	cP	Q	E	L	L	C		L	Y	p	P	W	D	K	N	F	T	E
FIX	K	-	-	V	D	A	F	C		E	T	-	-	-	-	-	-	-	-	G
FX	E	E	-	N	E	G	F	C		Y	Q	-	-	-	-	-	-	-	-	A
PC	S	K	-	K	K	L	A	C		D	E	-	-	-	-	-	-	-	-	S

Chg	95	96	97	98	a	b	99		145	147	148	a	b	c	d	e	f	149
FII	N	W	R	E	N	-	L		K	E	T	W	T	A	N	V	G	K
FIX	N	A	A	I	N	K	Y		F	H	K	-	-	-	-	-	-	G
FX	T	K	E	T	-	-	Y		H	E	K	-	-	-	-	-	-	G
PC	S	K	S	T	-	-	T		S	S	R	E	K	E	A	K	-	R

Chg	172	173	174	175	176		189	190	191	192	193	194		214	215	216	217	219	220
FII	T	R	I	R	I		D	S	C	Q	G	D		S	W	G	E	G	C
FIX	T	K	F	T	I		D	A	C	Q	G	D		S	W	G	E	E	C
FX	S	S	F	I	I		D	A	C	E	G	D		S	W	G	E	G	C
PC	M	S	N	M	V		D	A	C	E	G	D		S	W	G	E	G	C

lar elements. Thus, prothrombin, factor X(a), factor IX(a) and protein C all possess an amino-terminal g-carboxyglutamic acid rich domain (Gla). In the absence of calcium, this domain is loosely ordered (17–19), but folds in the presence of calcium to expose hydrophobic residues (20,21) which may function in anchoring these vitamin K dependent proteinases to phospholipid membranes (22). The Gla and catalytic domains of prothrombin are linked by two disulphide-knotted kringle domains, which become separated from the catalytic domain upon activation cleavage to form a-thrombin. In contrast, factor Xa, factor IXa and protein C each contain two epidermal growth factor (EGF)-like domains in place of the kringle domains, which are covalently linked to the catalytic domain and allow the activated proteinases to remain associated with the membrane. This confines coagulation to the site of injury, and greatly enhances the encounter frequency of enzymes and substrates.

The haemostatic proteinases circulate in the blood as inactive zymogen precursors. Activation occurs through the specific cleavage of the Arg15-Xxx16 bond, where Xxx is a conserved branched aliphatic residue (usually Ile). As a result of this cleavage, the new free amino terminus of (Ile)16 inserts into the body of the serine proteinase (the activation domain (23)), where it forms a buried salt bridge with the conserved Asp194 (immediately preceding the active site residue Ser195), causing the transition from zymogen to active serine proteinase. The haemostatic proteinases vary considerably in their activities, however; for example, factor IXa is virtually inactive in the absence of its cofactor VIIIa (24). Indeed, the influence of such cofactors is a striking feature of the coagulation, anticoagulation and fibrinolytic cascades. Thus, the activation of prothrombin by factor Xa is enhanced greatly in the presence of factor Va (3,25), as is the inactivation of factors Va and VIIIa by protein C in the presence of protein S (4). Particularly striking is the switching of thrombin specificity in the presence of thrombomodulin from the procoagulatory conversion of fibrinogen to fibrin to the anticoagulatory activation of protein C. Such a cofactor

requirement allows fine control of the cascades, regulating the activity of a particular factor at various stages of clot formation and degradation.

Thrombin structure and allostery

The actions of α -thrombin are central to haemostasis (1,9). Thrombin is a versatile enzyme, exhibiting both pro- and anti-coagulatory effects: the former through its conversion of fibrinogen to fibrin, its activation of the platelet thrombin receptor (26) and several coagulation cofactors, the latter via the thrombomodulin-mediated activation of protein C (4). Physiologically, its activity is regulated by serpins such as antithrombin III, heparin cofactor II and protease nexin I as well as the general proteinase scavenger α_2 -macroglobulin (27). Inhibition by antithrombin III and heparin cofactor II is strongly accelerated by the acidic glycosaminoglycan heparin. Thrombin is inhibited only weakly by other typical serine proteinase inhibitors such as the Kunitz inhibitor BPTI (28).

Despite its many substrates, thrombin exhibits a narrow specificity in its cleavage sites compared to the related enzyme trypsin. The thrombin molecule exhibits the characteristic fold of the trypsin-like serine proteinases (6,7) (Fig. 1). As a result of several characteristic insertions (see Table 1), the active site cleft of thrombin is both deeper and narrower than that of trypsin. Two pronounced insertion loops that border the active site cleft may play a major part in thrombin's restricted specificity: the '60-insertion loop', containing residue Trp60D, and the '148-insertion loop', containing residue Trp148 (Fig. 1). Comparison of structures of thrombin in complex with a wide variety of substrates and inhibitors indicates that the 148-loop can adopt a range of conformations (8,29); the 60-loop, on the other hand, appears to possess a rigid structure that differs less than 1.5 Å in nearly all structures determined to date (30).

A particularly striking feature of the thrombin molecule is its pronounced charge distribution; positively charged

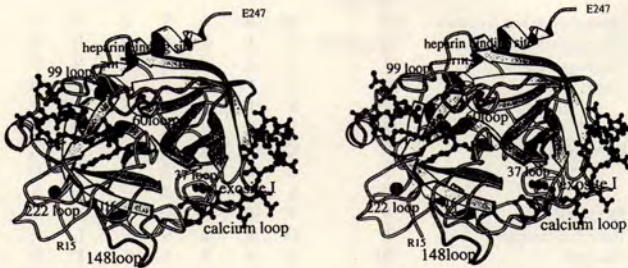


Fig. 1. Stereo view of human α -thrombin. This view defines the "standard" orientation, with the active-site cleft of the B-chain running from left to right ("west-east"), the direction in which a substrate polypeptide chain would run. The positions of various loop structures are indicated (see also table 1). The "60-insertion loop", in particular Trp60D, partially occludes the active site to the "north". To the "east", fibrinopeptide A occupies the active site, with the side chain of Arg16 entering the S1 specificity pocket; Phe8 occupies the aryl-binding site [34]. To the "west", the C-terminal acidic peptide of hirudin occupies the fibrinogen recognition site. This region corresponds to the 'calcium binding loop' of factor Xa, factor IXa and protein C. The sphere at the centre of the 222-loop represents the putative sodium binding site [56].

residues cluster to the east and north-west of the active site (Fig. 2), forming the fibrinogen recognition exosite and the heparin binding site, respectively. Haematophages such as the medicinal leech *Hirudo medicinalis*, the assassin bug *Rhodnius prolixus* and the soft tick *Ornithodoros moubata* all possess potent thrombin inhibitors that accommodate themselves to the restricted active site with concomitant binding of acidic residues to the basic fibrinogen recognition exosite (31–33) (Fig. 1).

The interaction of thrombin with fibrinogen provides a prime example of specific substrate recognition (34–36). Thrombin possesses a deep S1 pocket, with the acidic Asp189 at its base poised to make a salt bridge with the P1 arginyl residue of fibrinopeptide A. The conjuncture of hydrophobic residues provided by the 60-loop and the base of the cleft form the apolar binding site of thrombin, subdivided into the S2 pocket and the aryl-binding site. Fibrinopeptide A (Asp-Phe-Leu-Ala-Glu-Gly-Gly-Val-Arg) folds into a compact amphiphilic conformation (Fig. 1) such that hydrophobic residues cluster to form a single surface. These slot into the apolar binding site of thrombin; the P2 Val fills the S2 site, whilst the (formally P9 and P8) Phe and Leu residues occupy the aryl-binding site. Binding is driven by the complementary burial of lipophilic surfaces; the relatively small number of polar interactions serve to anchor and / or orient the peptide correctly in the hydrophobic environment.

Data have recently become available related to the primed site specificity of thrombin, including the crystallographic determination of the P1'-P3' sites for hirulog3 (37) and the fibrinogen A α chain (38). Antithrombin Denver, which has Leu in P1' instead of Ser, does not inhibit thrombin (39,40). Substitution of the P1' residue in antithrombin with each one

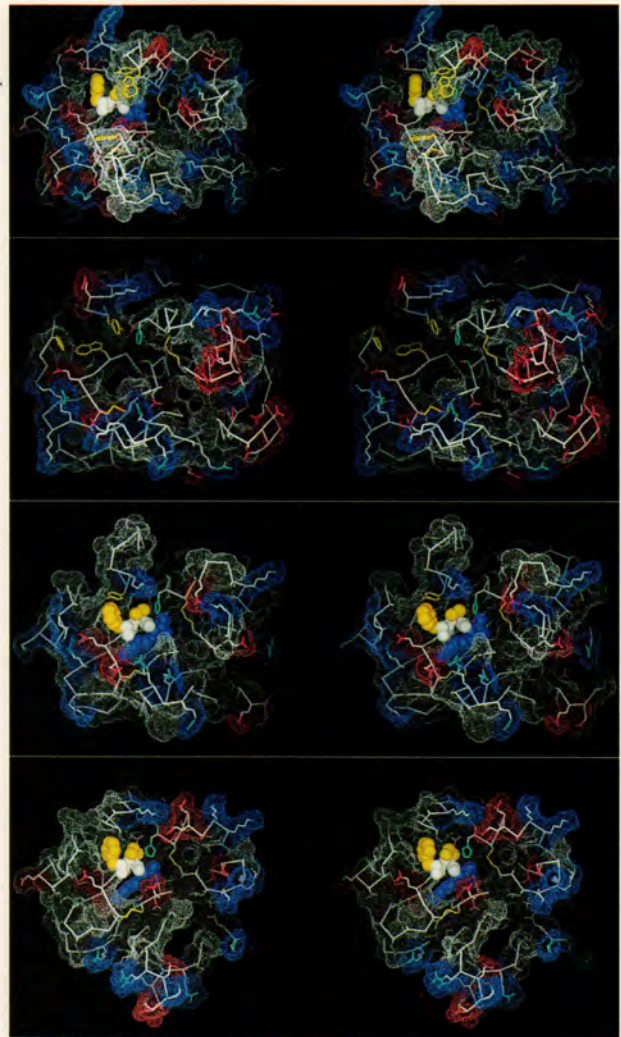


Fig. 2. Stereo views of the active site clefts of the haemostatic proteinases in standard orientation (see Fig. 1), displayed with Connolly dot surfaces; molecular surfaces are coloured dark blue, red, or white if formed by basic, acidic or other residues, respectively. Only the front parts of the molecules and of their molecular surfaces are displayed.

- The thrombin molecule with PPACK bound in the active site (see also Fig. 1) (6,7). The PPACK molecule snuggles under the 60-insertion loop, which partially occludes the active site to the north. Pronounced basic patches to the east and the north-west define the fibrinogen recognition exosite and the heparin binding site, respectively.
- The factor Xa molecule (see also Fig. 3) (10). The active site is unoccupied. Note the pronounced hydrophobic box to the north-west of the active site, and the clustering of acidic residues to the east (corresponding to the fibrinogen recognition exosite of thrombin). This latter site may play a role in complex formation with TFPI (85).
- The factor IXa molecule with PPACK bound in the active site (see also Fig. 4) (12). The PPACK molecule is in a strained conformation, pushing out the side chain of Tyr99 of the extended 99-insertion loop.
- The protein C molecule with PPACK bound in the active site (13). The acidic residues Asp60 and Glu60A to the north of the active site could influence the binding of basic P2 substrate residues. A basic patch is found to the east in a position equivalent to the fibrinogen recognition exosite of thrombin.

of the 20 naturally occurring amino acids (41) revealed a strong preference for Ser and Ala, followed by Gly and Thr; bulky amino acids at the P1' position destroyed all anti-thrombin activity. A potent thrombin inhibitor could be derived from the leech derived tryptase inhibitor (42) via a series of mutations that included the substitution of the P1' Ile residue for Ala. This preference for small polar P1' residues is due to Lys60F, which reaches into the active site cleft and partially obstructs the S1' site. Mutation of Lys60F to the shorter Glu (43) or Ala (44) results in reduced fibrinogen-clotting activity (P1' Gly, Table 1) and enhanced protein C-activating activity (P1' Leu). However, the P1' Gly of fibrinogen Aa does not fill this pocket completely (38), explaining the reduced inhibitory performance of the antithrombin mutant (P1'Ser→Gly) (41).

The P2' and P3' specificities of thrombin have been studied comprehensively using fluorescence-quenched substrates (45). Bulky hydrophobic residues in the P2' position (in particular Phe) are preferred to polar groups, and acidic side chains (Asp, Glu) are detrimental to catalysis, consistent with the crystallographic evidence which shows that the P2' site is bordered by the side chains of Leu40 and Leu41 (38). Less differentiation is seen in the P3' position, with a weak preference for basic side chains and an aversion for acidic residues. Site-directed mutagenesis shows that the intolerance for acidic P2' and P3' residues is a result of charge repulsion with Glu192 (46) and Glu39 of the 37-loop (44,45,47), involved in a charge network with Arg35 and Lys36 (7,34). These residues are also responsible for the positive contribution of basic P3' residues to substrate cleavage.

The P3-P3' sequence (Phe)-Pro-Arg-Ser-Phe-Arg should therefore be optimal for thrombin cleavage, allowing classification of natural substrates into 'near-optimal' and 'non-optimal' (45) (Table 2). Many of the 'non-optimal' cleavage sites (e.g. fibrinogen Aa and Bb, two of the factor V activation sites, protein C, heparin cofactor II) utilize binding at sites removed from the active centre: the fibrinogen recognition exosite and/or the heparin binding site. Thus, the cleavage of fibrinogen by thrombin requires an intact fibrinogen recognition exosite and the presence of an acidic region in the neighbourhood of the N-terminal disulphide knot. Similarly, the highly unfavourable P1 Leu residue of heparin cofactor II (48,49) is only accommodated in the presence of heparin, which allows simultaneous interactions at the fibrinogen recognition exosite and at the heparin binding site. Heparin cofactor II also possesses a heparin binding site, which appears to be occupied in the unliganded state by its acidic N-terminal peptide (50,51). The addition of heparin has two effects: it facilitates the binding to thrombin by linking the heparin binding sites of both inhibitor and enzyme, and it displaces the acidic peptide, making it available for interactions with the fibrinogen recognition exosite of thrombin (52). This complex mechanism allows accommodation of the unfavourable P1 Leu residue, the presence of which ensures that of all the haemostatic proteinases, only thrombin is inhibited.

Thus far, we have considered the actions of thrombin in terms of a static molecular model. Evidence for allosteric regulation of thrombin has accumulated over the last years,

Table 2. Principal substrate cleavage sites of the haemostatic proteinases. cmk: chloromethyl ketone

	P4	P3	P2	P1	P1'	P2'	P3'	P4'
Thrombin								
Optimal cleavage sequence	(F)	(F)	P	R	S	F	R	?
Fibrinogen Aa	⁷ DFLAEGG	G	V	R	G	P	R	V
Fibrinogen Bβ	¹¹ F	S	A	R	G	H	R	P
Factor V	⁷⁰ L	G	I	R	S	F	R	N
	¹⁰¹ L	S	P	R	T	F	H	P
	¹⁵⁴ W	Y	L	R	S	N	N	G
Factor VIII	³⁶⁹ I	Q	I	R	S	V	A	K
	⁷³⁷ I	E	P	R	S	F	S	Q
	¹⁶⁸ Q	S	P	R	S	F	Q	K
Factor XIII	³⁵ V	V	P	R	G	V	N	L
Thrombin receptor	³⁸ L	D	P	R	S	F	L	L
Protein C	¹⁶⁶ V	D	P	R	L	I	D	G
Antithrombin III	³⁹ Q	A	G	R	S	L	N	P
Heparin cofactor II	⁴⁴¹ F	M	P	L	S	S	T	Q
Protease nexin I	³⁴² L	I	A	R	S	S	P	P
		DF	P	R	cmk			
Factor Xa								
Prothrombin	³¹⁷ I	D	G	R	I	V	E	G
	²⁶⁸ I	E	G	R	T	A	T	S
Factor VII	¹⁴⁹ P	Q	G	R	I	V	G	G
Antithrombin III	³⁹ Q	A	G	R	S	L	N	P
TFPI domain II	¹⁰⁴ G	I	C	R	G	Y	I	T
	dansyl	E	G	R	cmk			
Factor IXa								
human Factor X	¹⁹⁴ N	L	T	R	I	V	G	G
bovine Factor X	¹⁹³ Q	V	V	R	I	V	G	G
chicken Factor X	²⁰⁰ V	D	T	R	I	V	G	G
Protein C								
Factor Va	³⁰³ K	K	T	R	N	L	K	K
Factor Va	⁵⁰³ L	D	R	R	G	I	Q	R
Factor Va	⁶⁷⁶ M	A	T	R	K	M	H	D
Factor V	⁹⁹¹ R	L	K	R	S	Q	F	L
Factor VIIIa	³³³ P	Q	L	R	M	K	N	N
Factor VIIIa	⁵⁵⁹ V	D	Q	R	G	N	Q	I

¹ P1 denotes the substrate residue immediately N-terminal to the cleavable bond, P2 the preceding residue etc. Similarly, P1', P2' etc. denote the first, second etc. residues C-terminal to the cleavage site. S1, S2, ..., S1', S2', ... are the sites on the proteinase that receive the P1, P2, ..., P1', P2', ... residues.

however -reviewed in (53). Two allosteric states have been described, the 'slow' and 'fast' forms (54); transition to the fast form is triggered by the specific binding of a Na⁺ ion (54,55). The two forms have different relative activities towards macromolecular substrates: the slow form activates protein C preferentially, whilst the fast form cleaves fibrinogen. A recent crystallographic study of Rb⁺-thrombin (56) reveals a putative Na⁺ binding site, with the cation coordinated by the carbonyl oxygen atoms of Tyr184A, Arg221A and Lys224 (Fig. 1) and stabilised by the negative charges of Asp221 and Asp222.

Interactions in the vicinity of the 222-loop play a role in the modulation of thrombin's activity. Mutation of the nearby Glu217 to Ala shifts the specificity of thrombin in favour of protein C activation (57), which may serve to alleviate electrostatic repulsion with the non-optimal P3 Asp residue of protein C. Arg221A forms a salt bridge with Glu146 of the 148-loop; the loss of this contact in the genetic variant thrombin Salakta, where residue 146 is mutated to Ala (58) results in reduced catalytic activity. Similarly, the activity of a thrombin mutant lacking residues Glu146, Thr147 and Trp148 is severely compromised (59). The exceedingly long 148-loop exhibits considerable conformational variability (8,29). Resistance to exogenous proteolytic cleavage of the

148-loop is conferred by the binding of active site specific inhibitors of thrombin, and by the exosite binding of the C-terminal acidic peptide of hirudin (60). This suggests that the 148-loop could mediate cross-talk between the active site and the fibrinogen recognition exosite, possibly through the Glu146-Arg221A salt bridge, providing a structural explanation for the concept of allosteric linkage (61,62).

What is the structural origin of this allosteric regulation of activity? Analyses of thrombin-ligand complexes reveal that the rim of the S1 pocket can vary in its conformation (30,63–65). In particular, the segment Cys191-Glu192-Gly193 may 'collapse' to make closer contacts to the P1 residue; in the case of a substrate, this could clamp the cleavable bond into a position optimal for cleavage. Through the disulphide bridge Cys191-Cys220, this segment is linked to the 222-loop, which in turn is involved in Na⁺ binding and linked to the fibrinogen recognition exosite via the 148-loop. The 222-loop acts as an 'effector', while the 148-loop behaves as a 'reporter' of exosite binding.

The activation of protein C is an important regulator of thrombotic activity (4). In the presence of thrombomodulin, protein C becomes the major substrate for thrombin; in its absence, however, protein C is a poor substrate. With its acidic residues at P3 and P3' and its bulky P1' Leu residue, protein C probably represents the least favourable thrombin substrate. Thrombin mutants E192Q (46) and E39K (47) activate protein C 22 and 2.2 times more rapidly than wild-type thrombin, respectively, suggesting that thrombomodulin binding induces a movement of Glu192 and/or Glu39, thereby switching thrombin's specificity (34). We have recently shown that a bulky P1' Ile residue can be accommodated in the restrictive active site of thrombin upon favourable binding at the fibrinogen recognition exosite (42), indicating that thrombin can exhibit a degree of plasticity. Indeed, the thrombomodulin dependent activation of protein C results in conformational changes in the thrombin molecule (66).

Residue 192, Glu in native thrombin, has been shown to be an important determinant for ligand specificity in thrombin (46,67), factor Xa (68), factor VIIa (69) and protein C (70). Thrombin residue Glu192 is solvent accessible, and is therefore rather flexible (8). In addition to repelling acidic P3 and/or P3' residues (46), the negative charge on the side chain would hamper 'canonical' complex formation with an inhibitor or substrate due to repulsion of the P2 and P3 main-chain carbonyl groups. Although a reorientation of this side chain has been correlated to exosite binding (63–65), recent biochemical data show that such a reorientation is in itself insufficient to mitigate unfavourable interactions with acidic P3 residues (71). Glu192 is utilised to great effect by the highly specific thrombin inhibitor rhodniin from *Rhodnius prolixus* (32,72). In P1, rhodniin possesses the unusual His residue, which allows simultaneous hydrogen/ionic bonds to be made with both Asp189 and Glu192.

Glu192 has a profound effect on the interaction of thrombin with the paradigmatic serine proteinase inhibitor BPTI. Docking of the inhibitor into the active site of thrombin indicated severe steric clashes with the 60-loop on complex formation (7), consistent with its poor inhibition (28). Deletion of the 60-insertion loop (thrombin desPPW) results in

nanomolar inhibition by both BPTI (73) and TFPI (67), while deletion of the 148-loop (thrombin desETW) results in a modest 25-fold decrease in K_i for BPTI compared to wild-type thrombin (59). Quite unexpectedly, however, the single atom mutation E192Q in thrombin allowed the binding of BPTI and TFPI, with K_i values in the 10⁻⁸ M range (67).

The structure of the thrombin mutant E192Q in complex with BPTI (74) reveals complex rearrangements of the surface loops surrounding the active site cleft. The most striking feature of the complex is a large reorganisation of the 60-loop: at the apex of the loop, the C_α atom of Trp60D is displaced by some 8 Å. Displacement is not a simple rigid body movement; the main chain conformation is altered significantly, indicating that major disruptions of the loop occur upon binding to BPTI. The reorganisation of the 60-loop, stabilised by a number of (largely van der Waal's) contacts to the inhibitor molecule, causes an additional remodelling of the 37-loop, and thus an effect on the fibrinogen recognition exosite. It is therefore conceivable that the reverse process could occur: that strong binding at the fibrinogen recognition exosite could facilitate partial expulsion of the 60-loop.

Such a structural rearrangement of thrombin may be important in its interaction with the serpin antithrombin. Serpins exhibit a complex inhibition mechanism, in which an initial loose association of the proteinase and serpin evolves into a tight irreversible complex (75). Together with the results of recent experiments using thrombin mutants (76,77), the 'open' thrombin structure suggests that its interaction with antithrombin could occur as follows: (i) approach of antithrombin to thrombin, (ii) expulsion of thrombin occluding loops, (iii) canonical complex formation, stabilisation of thrombin loops, and (iv) transition to stable, non-reversible complex. The addition of heparin as a template, linking the enzyme and inhibitor components, could provide sufficient binding energy to counterbalance the energetically unfavourable 60-loop expulsion.

The opening of the active site cleft is attained only at a considerable energetic cost (see (74)), which could be supplied by suitable binding at the fibrinogen recognition exosite or the heparin binding site. Exosite binding can achieve three (maybe interdependent) possible results: the provision of novel interaction surfaces, the exertion of allosteric activation effects, or the opening of the active site cleft to receive bulky substrates or inhibitors. Allosteric changes of this nature could explain the lack of fibrinogenolytic activity of the membrane-bound active thrombin intermediate meizothrombin (78–80), whose heparin binding exosite is occupied by prothrombin kringle F2 (81), but whose fibrinogen recognition exosite is free (82–84).

Factor Xa

The catalytic domain of factor Xa exhibits an open active site cleft similar to that of trypsin or chymotrypsin (10) (Fig. 3), with a distinctive pattern of charged residues of non-uniform distribution (Fig. 2). A basic site to the north-west of the active site maps to the heparin binding site of thrombin. In contrast to thrombin, however, this site does not appear to play a role in heparin mediated inhibition by antithrombin

III, and presumably has another function. An acidic patch occurs to the east of the active site (the 39-loop), corresponding to the basic fibrinogen recognition exosite of thrombin.

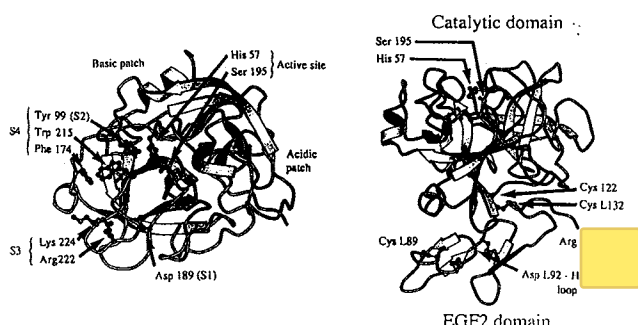


Fig. 3. The factor Xa molecule

(a) The catalytic domain of factor Xa in standard orientation (10).

The EGF domain sits "behind" the molecule (see Fig. 3b). Active site residues Ser 195, His 57 and S1 residue Asp 189 are shown. The putative S4 site (the 'hydrophobic box', bounded by Phe 174, Trp 215 and Tyr 99) is shown, as are residues Arg 222 and Lys 224 (putative S3 site for prothrombin).

(b) View of factor Xa from the 'south'; the molecule has been rotated by 90° from the standard orientation about a horizontal axis, such that the active site runs from left to right at the top of the figure. The second EGF-like domain is seen at the bottom of the figure; the first EGF domain, which would be linked to Cys L89, is flexible in the crystal forms examined (10,11). The three disulphides form a ladder through the core of the domain, holding together the short β -sheets.

The cigar-shaped second EGF domain (EGF2) of the light chain is found at the base of the molecule diametrically opposed to the active site cleft, disulphide linked to the catalytic domain (Fig. 3b). The intimate contacts between the catalytic domain and EGF2 suggest that the two operate as a single unit. In contrast, the link between EGF1 and EGF2 is probably flexible, as no density could be detected for EGF1 in any of the crystal forms examined (10,11).

The specificity pocket of factor Xa, with the acidic residue Asp189 at its base, closely resembles that of thrombin, explaining the preference of factor Xa for P1 arginyl residues. **Particularly striking is the presence of the 'hydrophobic box' formed by residues Tyr99, Phe174 and Trp215.** The side chain of Tyr99 would restrict P2 residues of substrates and inhibitors to glycine, whilst the hydrophobic box would be ready to accept bulky hydrophobic P4 residues. An acidic residue at P3 could interact with the basic side chains of Arg222, Lys223 and Lys224. The cleavage sequences of human prothrombin (Ile-Glu/Asp-Gly-Arg-) (Table 2) would therefore appear to be optimally suited to the factor Xa active site. Some factor Xa substrates possess larger P2 residues, however; in this case, it is expected that the phenolic side chain of Tyr99 would swing out, as proposed for the factor Xa: tissue factor pathway inhibitor (TFPI) interaction (85) (see below).

Structural studies on the specific synthetic factor Xa inhibitor DX9065a have identified an electronegative cavity formed by the carbonyl oxygens of residues Lys96 (11) or

Thr98 (86) and Glu97 and the carboxylate side chain of the latter. Thus, in addition to accepting hydrophobic P4 residues, this 'nitrogen cation hole' may allow binding of positively charged P4 residues, and may be of use in the design of factor Xa specific inhibitors (87). Such a weak ionic interaction may play a role in the formation of the proposed factor Xa:TAP (tick anticoagulant peptide) complex (33).

Extrinsic generation of factor Xa is rapidly extinguished by tissue factor pathway inhibitor (TFPI) (reviewed in (88)), whose second **Kunitz-like domain inhibits factor Xa**. Structural studies of TFPI-II (85) suggest that inhibition of factor Xa causes disruption of the hydrophobic box, with the side chain of Tyr99 swinging out towards Phe174. The resulting increased hydrophobic surface area is then buried by aliphatic residues of TFPI-II. Electrostatic interactions appear to play a significant role in the formation of a putative factor Xa:TFPI-II complex, with basic groups of TFPI-II approaching the acidic 39-loop of factor Xa and an acidic patch on the opposite surface of TFPI-II orientated towards the basic 148-loop (85) (see Fig. 2).

Factor IXa

The catalytic domain of factor IXa (12) most closely resembles that of factor Xa (Fig. 2), including the conservation of residues Tyr99, Phe174 and Trp215 that make up the hydrophobic box in factor Xa. Factor IXa exhibits an insertion of two residues in the 99-loop, however (table 1), so that the S4 pocket is probably barely accessible in the native structure. Although this site is occupied in the crystals by the phenylalanine moiety of PPACK, the strained conformation of the covalently bound inhibitor (Fig. 2) suggests that it has perturbed the native structure. Restricted access to the S4 pocket by the 99-insertion loop, and to the S2 pocket by the side chain of Tyr99 as in factor Xa, could explain the (surprising) lack of conservation of P4-P2 residues in the activation sequences of factor X from different species (Table 2).

Of particular note is residue Glu219 at the entrance to the S1 pocket, a residue that is Gly in virtually all other trypsin-like serine proteinases. In factor IXa, Glu219 adopts a high-energy main-chain conformation normally only available to glycyl residues, suggesting that the structural integrity of the S1 pocket is compromised. Molecular dynamic simulations suggest a partial collapse of the S1 pocket upon elimination of PPACK (12). This provides a plausible explanation for the poor intrinsic activity of factor IXa towards chromogenic substrates (89,90), and suggests that factor IXa is held in a zymogen-like state.

As in factor Xa, the catalytic domain of factor IXa makes intimate contacts with its EGF2 domain, with a common overall arrangement and aromatic-rich interface. The EGF1 domain of factor IXa is defined, giving the molecule the appearance of a tulip (Fig. 4). The interface between the two EGF domains is reminiscent of a ball-and-socket joint, with a convex hydrophobic loop of EGF1 set into a concave hydrophobic socket of EGF2. Rotation of the joint appears to be restricted by the shortness of the peptide linking the two

domains (5 residues as opposed to 7 residues in factor Xa) and by conserved ionic contacts. Due to a lack of calcium in the crystals, the orientation of the Gla domain is defined only by weak electron density; recent NMR and small-angle X-ray studies show that the relative orientation of the Gla and EGF1 domains changes upon calcium binding (91).

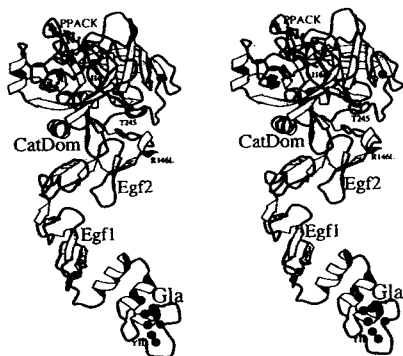


Fig. 4. Stereo view of the factor IXa molecule from the 'south' (in a similar orientation to Fig. 3b) (12). The covalently bound PPACK molecule (top) indicates the location of the active site. The elongated molecule resembles a tulip, with the catalytic domain as flower, the two EGF domains resembling the stalk, and the Gla domain the bulb. Spheres represent the putative locations of calcium ions in the catalytic, EGF1 and Gla domains. Surface mutations related to haemophilia B are located at the front and towards the left of the molecule in this view.

Hereditary deficiency of factor IXa is responsible for the X-linked bleeding disorder haemophilia B. Mapping of known haemophilic variants of high antigenic level reveals an 'haemophilic surface' that extends along the entire concave surface of the molecule (at the front of Fig. 4) (12). Presumably, this surface is involved in the interaction of factor IXa with its cofactor, factor VIIIa. All four modules of the molecule are involved in contacts; in the catalytic domain, a cluster of mutations is seen in the activation domain and in the vicinity of the 'intermediate helix' (residues 162-170) (92-94) (to the left, Fig. 4). The equivalent helix in factor VIIa is involved in contacts to tissue factor (14).

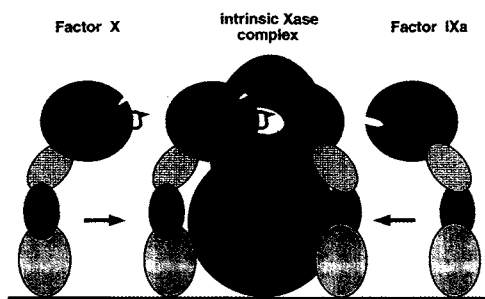


Fig. 5. Schematic drawing of intrinsic Xase complex formation. In the absence of factor VIIIa, the specificity pocket of factor IXa is mobile and incompletely formed. The bent factor IXab molecule could contact factor VIIIa along one side. Factor X, of a similar overall shape, binds to factor VIIIa from the opposite side; the interaction forces the factor X activation site Arg194-Ile195 into the factor IXab active site, opening the S1 specificity pocket and enabling activation cleavage.

The structure of factor IXa and the distribution of mutations therefore imply an extended interaction surface between factor IXa and factor VIIIa, and suggest a working model for intrinsic Xase complex formation (Fig. 5). Factor VIIIa could partially stabilise the activation domain and the S4 site; however, the lack of activity of intrinsic tenase towards chromogenic substrates suggest that full catalytic activity is only achieved in the presence of factor X, the physiological substrate. This substrate-assisted activation provides a particularly acute safety mechanism in the coagulation process.

Protein C

The catalytic domain of activated protein C (aPC) most closely resembles that of thrombin. A prominent cluster of basic residues (the anion binding exosite) is located in the same region where thrombin has its fibrinogen recognition exosite (Fig. 2). By analogy, this cluster may form a secondary binding site for its substrates, factors Va and VIIIa, which possess highly anionic regions C-terminal to their inactivation cleavage sites. Where thrombin has its heparin binding site, however, aPC has a surface dominated by small polar but uncharged residues, providing resistance to antithrombin III. Although the serpin PCI (Protein C Inhibitor) binds heparin and appears to function via a ternary complex mechanism (95), the heparin binding sites of PCI and antithrombin III are topologically distinct, raising the possibility that heparin bridges between the protein C anionic binding site and the PCI heparin binding site (96). It has also been suggested that the 148-insertion loop, which is only one residue shorter than that in thrombin (table I), may play a role in protein C inhibition by the serpin α_1 -proteinase inhibitor (97).

The active site cleft of aPC is rather polar, and is open and accessible to macromolecular substrates. The acidic residue Glu60A, which should not hinder approach to the active site cleft, may help stabilize the Arg residue in the P2 position of factor Va at the Arg 506 cleavage site. The central position of Glu60A suggests that it could also affect the binding of basic residues at P4, P1' and P3' (see Table 2, Fig. 6). The P2 pocket, relatively open due to the short Thr99 side chain, is particularly suited to large polar residues (see Fig. 2). The polar nature of the S4 pocket, flanked by Thr 99 and Asn 174 and restricted in size by the side chain of Met175, probably explains the decreased affinity for PPACK compared to thrombin with its aryl-binding site. The primed side of the active-site groove displays an unusually large P3' pocket, unique to aPC, caused by the presence of an Ala residue at position 41; this pocket is closed in thrombin by Leu41, and in factors Xa and IXa by Phe41. The side chain of Leu40, which reaches towards the S2' site, could cause a substrate main chain to follow a slightly different course at P2' compared to the canonical conformation. Similar to thrombin, protein C possesses the acidic Glu192 residue at the entrance to the active site cleft, which confers resistance to plasma protease inhibitors (70), and may be a cause for its remarkably long *in vivo* half-life (approximately 15 minutes).

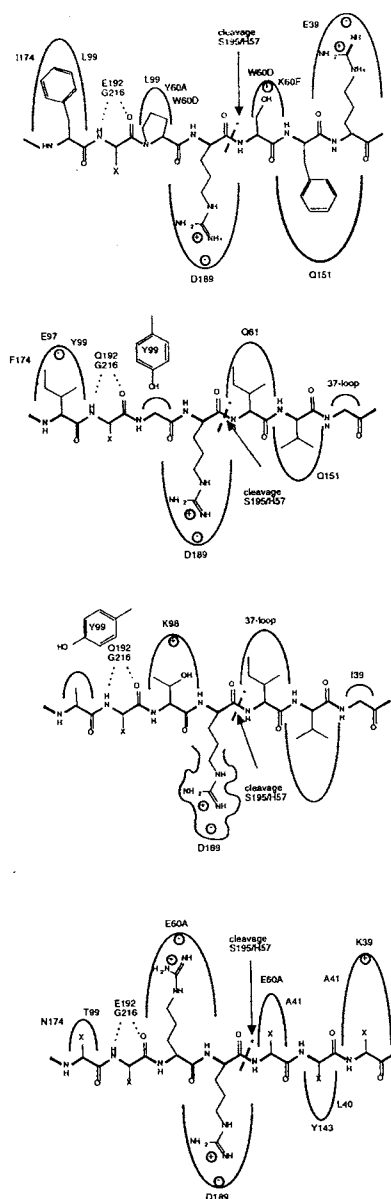


Fig. 6. Schematic diagram of P4-P3' specificity determinants of haemostatic proteinases.

- Thrombin. The 60-insertion loop places stark constraints on substrate/inhibitor P2, P4 and P1' residues. The S2 site is best suited to small hydrophobic residues, the S4 site to aromatic groups, and the S1' to small polar side chains.
- Factor Xa. The side chain of Tyr99 restricts substrate P2 residues to Gly, while the S4 site is suitable for bulky hydrophobic and basic residues. The putative prime sites appear to be suited to Ile-Val-Gly sequences.
- Factor IXa. The S1 site seems to exist in a zymogen-like state that becomes ordered in the presence of factor VIIIa and factor X. The S4 site may be restricted by the 99-insertion loop, while the S2 site is relatively open and polar. Like factor Xa, the primed sites should be appropriate for the activation of other serine proteinases.
- Protein C. In contrast to the other serine proteinases described here, the specificity sites of protein C are rather polar. In particular, the S2 and S1' sites may prefer basic residues. The S4 site appears restricted, while the S3' site is remarkably open.

Discussion

The structural studies presented here reveal how the common serine proteinase scaffold has been utilized in the evolution of the highly complex process of haemostasis. Subtle variations in the positioning of insertions and charged and lipophilic residues bordering the active site cleft result in a myriad constellation of sites that allow large variations in specificity for substrates and inhibitors. Combining the available structural and biochemical data, the substrate specificity sites of thrombin, factor Xa, factor IXa and protein C are summarized in Fig. 6. These specificities are masked by a variety of other effects, however, such as differing relative activities in the presence of effectors or cofactors.

The S1-S1'-S2'-S3' sites of factors Xa and IXa appear to be especially suited to the cleavage of Arg-Ile-Val-Gly sequences, i.e. they are ideal for the activation of other serine proteinases. In contrast, thrombin and protein C are particularly unsuited to serine proteinase activation, due to their restricted S1' and large polar S3' sites. This reflects the roles of factors Xa and IXa as amplifiers and thrombin and protein C as end-points of the proteolytic cascades. The absence of activatory potential of thrombin and protein C and the inactivity of factor IXa in the absence of its cofactor VIIIa helps guard against unwanted initiation of proteolytic cascades in the vasculature at sites removed from that of injury. Nevertheless, the intrinsic activity of factor Xa represents a possible risk of coagulation cascade generation, so that its inhibition by antithrombin or TFPI should be rapid and complete.

Although thrombin is a poor proteinase activator, thrombin catalyzes the activation of protein C in the presence of thrombomodulin. The exosite binding afforded by thrombomodulin may provide novel substrate binding sites, but can also regulate thrombin's catalytic activity allosterically. We are just beginning to understand the structural nature of such allosteric regulation, which may be through the opening of the active site cleft and / or subtle alterations in the activation domain linked to exosite binding.

It is conceivable that the thrombin molecule (and by analogy the other haemostatic proteinases) undergoes progressive modulation of its activity and binding sites during the course of clot formation. Thus, the inactive zymogen precursor prothrombin is converted to the membrane-bound intermediate meizothrombin, the heparin binding site of which is occupied by prothrombin kringle F2; not only would this protect against antithrombin inhibition, this may result in slightly different macromolecular substrate specificities and activity. Similarly, a-thrombin, thrombomodulin-bound and fibrin-bound thrombin species might exhibit subtle gradations in their specificity and activity, further influenced by changes in the microenvironment such as the concentration of sodium or the presence of heparin.

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